Torn GA: diakoptics in a running evolution

Does tearing the circuit (Kron's diakoptics) with cached pieces make a value-genome GA faster, with the same result? — CLAUDE.md §3a, milestone M9b

kronspice simulation report · 2026-11-07 · population 40, 30 generations, seed 1

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Summary

A two-pole RC pre-filter (R1, C1, R2, C2, from the E24 and E12 series) is evolved in front of a fixed plant, joined to it by a resistor RT, for a 5 kHz cutoff at the plant's output. The same run (same seed) is made three times: AC analysis by modified nodal analysis (MNA), by Kron's node-pair form untorn, and by Kron's diakoptics torn at RT with the plant's factored pieces cached and shared by all workers. A mutation changes only the pre-filter, so only its piece is re-factored and the plant's factors are reused from one candidate to the next. This is the acceptance example of milestone M9b.

- **Same best design from all three solvers** on both plants; the program exits non-zero if they differ. On a 16-CPU run of the line: R1 13 Ω, C1 47 nF, R2 3 kΩ, C2 390 nF, fitness 5.6×10^{-6}.
- **Torn is an order of magnitude faster than untorn Kron:** 11.1× on the 40-section LC line and 5.6× on the 12×12 RC grid (two CPUs); 8.2× on the line with 16 CPUs.
- **Torn Kron is still slower than sparse MNA:** 2.0× on the line and 1.5× on the grid (two CPUs), 2.3× on the line with 16 CPUs. MNA stays the default; tearing is opt-in.
- **The cache works as intended:** on the line every candidate factors its own 4-unknown piece and reuses the 121-unknown line (233,046 pieces taken from the cache, 226,038 factored, 16-CPU run).
- **Torn equals MNA:** node voltages agree to 4.5×10^{-15} of the largest node voltage at 10 kHz, and the power taken by the R, L, C branches is identical in both (Roth's invariance).

What is being tested

CLAUDE.md §3a gives the reason *kron* exists in this project: a mutation usually changes one part of a circuit, so with tearing only the torn piece containing that change is re-solved and the other pieces' factored solutions are reused from the parent. M9b is done when the value-genome GA runs faster with tearing than without, with the same results.

The model

Circuit

V1 in 0 AC 1, the pre-filter R1 in m 1k, C1 m 0 10n, R2 m f 1k, C2 f 0 10n, the coupling RT f p0 10, and one of two plants from p0 to out:

- **line:** n sections of 1 Ω, 1 µH, 1 nF (√(L/C) ≈ 31.6 Ω), matched by RL out 0 31.6; 40 sections by default, 125 unknowns;
- **grid:** an n×n RC mesh, 10 Ω between neighbours and 1 nF to ground at every node, driven at one corner and loaded by 100 Ω at the other; a plant whose LU factorization fills in; 12×12 by default, 144 nodes.

Problem

Item	Setting
Genes	R1, R2: 10 Ω – 100 kΩ (E24); C1, C2: 100 pF – 10 µF (E12)
Objective	*evo.Cutoff* at node out, target 5 kHz
Analysis	.ac dec 50 100 1meg (201 frequencies per candidate)
Run	population 40, 30 generations (1,240 evaluations), seed 1

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Three solvers

- **MNA:** the ordinary *analysis.AC* path (sparse Gilbert–Peierls LU with a reused pattern, M6).
- **Kron, untorn:** *evo.Problem.Tear* with no torn branches, so every candidate goes through the node-pair form $Y' = A^T \cdot Y \cdot A$ (M9a).
- **Kron, torn at RT:** *kron.TornAC* cuts RT out, so the node-pair matrix is block-diagonal (pre-filter in one block, plant in the other) plus RT's rank-one term.

The torn solve, with B the block-diagonal matrix, U the torn branch's incidence, D its admittance and c the source vector:

```
z = B^-1·c                        each piece on its own
(I + D·U^T·B^-1·U)·w = D·U^T·z     interconnection system, order 1 here
x = z − B^-1·U·w
```

Here w is the current through RT, Kron's intersection current through the open path that tearing created. The torn solve refines against the full matrix and checks its componentwise backward error, reporting a nearly floating piece instead of returning a wrong answer. A *kron.TornCache* keeps factored pieces by their exact content (every entry's bit pattern, confirmed entry by entry on a hit), bounded by count and by 128 MiB; a cached and an uncached solve are bit-identical.

Results

Milestone run, two CPUs

Plant	MNA	Kron node form, untorn	Kron diakoptics, torn at RT	Same best design
40-section LC line (125 unknowns)	6.4 s	138.8 s	12.5 s (11.1× faster than untorn)	yes
12×12 RC grid (144 nodes)	21.3 s	177.0 s	31.5 s (5.6× faster than untorn)	yes

On the line the plant's pieces were taken from the cache 235,394 times and the pre-filter's piece factored 223,690 times (doc/M9b.md).

Line run on a 16-CPU machine (2026-11-01)

Solver	Time	Pieces cached / factored
MNA	1.18 s	
Kron node form, untorn	21.80 s	9 / 229,533
Kron diakoptics, torn at RT	**2.67 s**	**233,046 / 226,038**

Best design in all three: R1 13 Ω, C1 47 nF, R2 3 kΩ, C2 390 nF, fitness 5.6×10^{-6}. Torn is 8.2× faster than untorn Kron here.

Against MNA

The torn Kron solve is still slower than the MNA engine: 2.0× on the line and 1.5× on the grid in the two-CPU run, and 2.3× on the line in the 16-CPU run. Once the fixed piece is cached, what remains per frequency is building the primitive network, the branch relations and the node-pair rows, which costs about as much as MNA's whole sparse factorization of a chain. On the grid, where the LU fills in, the gap narrows. Reusing the per-network structure across the frequencies of one sweep, and building only the changed piece of a mutated candidate, are the steps listed in doc/M9b.md that would let the torn path overtake sparse MNA.

Torn solution against MNA (export sheet)

The 40-section line problem, analysed at 10 kHz, is torn at RT into 2 pieces of 4 and 121 unknowns joined by 1 link; the interconnection system has order 1, against 125 unknowns for the whole circuit.

Quantity	Torn (Kron)	Untorn (MNA)
Link current through RT	0.000454594 A, phase −20.73°	
Voltage across RT	0.00454594 V, phase −20.73°	
Power V·I* in the R, L, C branches	0.000553677, phase −16.21°	0.000553677, phase −16.21°
Node out	0.0143161, phase −28.08°	0.0143161, phase −28.08°

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Quantity	Torn (Kron)	Untorn (MNA)
Node f	0.0369572, phase −23.80°	0.0369572, phase −23.80°
Largest node-voltage difference	4.5×10^{-15} of the largest node voltage	

Validation

- *TestSameBestFasterTorn* runs a 20-section line (population 16, 6 generations, seed 3) and requires the same best values from all three solvers and a torn time below the untorn one.
- *BenchmarkGeneration* times one generation of 8 candidates with a fresh seed each iteration (the plant's pieces stay cached, the pre-filter's do not) for line/40 and grid/12, by MNA and torn.
- Milestone checks in doc/M9b.md: the torn solve equals the untorn solve and MNA on the linear golden circuits and 60 random circuits, 2,289 torn solves, worst 5.1×10^{-11} relative to MNA; power invariant under tearing; a cut that leaves a singular block is reported, not solved.

Limits

- Cuts are named by hand; there is no automatic partitioner in this example.
- Only AC analysis of linear candidates is torn; nonlinear circuits and circuits with idealized active elements are solved by MNA.
- Timings depend on the machine; the README reports a two-CPU and a 16-CPU run.

Reproducing the results

```
go run ./examples/torn_ga                      # 40-section LC line
go run ./examples/torn_ga -plant grid -n 12    # 12×12 RC mesh
go run ./examples/torn_ga -gens 30 -pop 40 -seed 1
go test ./examples/torn_ga                     # same best design, torn faster than untorn
go test ./examples/torn_ga -bench Generation   # one generation, fresh seeds
```

Sources: G. Kron, Diakoptics (1963); G. Kron, Tensor Analysis of Networks (1939), p. 133; H. H. Happ, Diakoptics and Networks (1971), Ch. XII; P. W. Aitchison, Quart. Appl. Math. 41, 265–272 (1983); J. P. Roth, Quart. Appl. Math. 17 (1959), Sec. 5; J. R. Gilbert, T. Peierls, SIAM J. Sci. Stat. Comput. 9, 862 (1988).

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